

Zach Perzan

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APPOINTMENTS

Assistant Professor Department of Geoscience University of Nevada, Las Vegas	2024 –
Research Fellow CarbonPlan	2026

EDUCATION

Ph.D. Earth System Science, Stanford University Gerald J. Lieberman Fellow NSF Graduate Research Fellow	2024
B.A. Geology (Honors), Middlebury College <i>Summa Cum Laude</i>	2015

HONORS & AWARDS

NSF Postdoctoral Fellowship, <i>National Science Foundation</i>	2023
Preparing Future Professors Fellowship, <i>Stanford University</i>	2022
Gerald J. Lieberman Fellowship, <i>Stanford University</i>	2021
Outstanding Paper, <i>Computational Methods in Water Resources</i>	2020
Outstanding Paper, <i>Stanford Deep Learning Symposium</i>	2019
NSF Graduate Research Fellowship, <i>National Science Foundation</i>	2017
National Award for Excellence in Research, <i>Council on Undergraduate Research</i>	2015
White Outstanding Research Award, <i>Middlebury College</i>	2015
Outstanding Paper, <i>Vermont Geological Society</i>	2014

PUBLICATIONS

Manuscripts under review (indicates student author)*

16. **Z. Perzan**, K. Boye, J. R. Bargar and K. Maher. Solute release from floodplains during distinct types of inundation events: Insights from field experiments and reactive transport simulations. [[link](#)]

Peer-reviewed articles

15. K. Maher, D.B. Rogers, **Z. Perzan** and T. Kukla. Surface proton buffering constrains alkalinity export from enhanced weathering, *Environmental Science & Technology*. [[link](#)]
14. A. Rasmussen, K. Langenfeld, B. Tolar, **Z. Perzan**, K. Maher, E. Cardarelli, J.R. Bargar, K. Boye and C. Francis (2025). Floodplain nitrifiers harbor the genetic potential for utilizing a wide range of organic nitrogen compounds, *mSystems*. [[link](#)]

13. L. Wang, T. Babey, **Z. Perzan**, S. Pierce, M. Briggs, K. Boye and K. Maher (2025). Quantifying groundwater response and uncertainty in beaver-influenced mountainous floodplains using machine learning-based model calibration, *Water Resources Research*. [\[link\]](#)
12. T. Dai*, K. Maher and **Z. Perzan** (2025). Machine learning surrogates for efficient hydrologic modeling: Insights from stochastic simulations of managed aquifer recharge, *Journal of Hydrology*. [\[link\]](#)
11. K. Maher and **Z. Perzan** (2024). Reactive transport as a scientific framework. In A. Shahar (Ed.), *Treatise on Geochemistry* (Third edition). Elsevier: Amsterdam. [\[link\]](#)
10. T. Babey, **Z. Perzan**, S. Pierce, D.B. Rodgers, L. Wang, R. Carroll, J.R. Bargar, K. Boye and K. Maher (2024). Mountainous floodplain connectivity in response to hydrological transitions, *Water Resources Research*. [\[link\]](#)
9. **Z. Perzan** and K. Maher (2024). Transport, dispersion and degradation of nonpoint source contaminants during flood managed aquifer recharge, *Vadose Zone Journal*. [\[link\]](#)
8. **Z. Perzan**, G. Osterman and K. Maher (2023). Controls on flood managed aquifer recharge through a heterogeneous vadose zone: hydrologic modeling at a site characterized with hydrogeophysics, *Hydrology and Earth System Sciences*. [\[link\]](#)
7. **Z. Perzan** and T. Chapin (2023). WellSTIC: A cost-effective sensor for point dilution tests to measure groundwater velocity in shallow aquifers, *Water Resources Research*. [\[link\]](#)
6. T. Babey, K. Boye, B. Tolar, M. Engel, et al. (2022). Simulation of anoxic lenses as exporters of reactivity in alluvial aquifer sediments, *Geochimica et Cosmochimica Acta*. [\[link\]](#)
5. **Z. Perzan**, T. Babey, J. Caers, et al. (2021). Local and global sensitivity analysis of a reactive transport model simulating floodplain redox cycling, *Water Resources Research*. [\[link\]](#)
4. Q. Li, L. Wang, **Z. Perzan**, J. Caers, et al. (2021). Global sensitivity analysis of a reactive transport model for mineral scale formation during hydraulic fracturing, *Environmental Engineering Science*. [\[link\]](#)
3. J. Damerow, C. Varadharajan, et al. (2021). Sample identifiers and metadata to support data management and reuse in multidisciplinary ecosystem sciences, *Data Science Journal*. [\[link\]](#)
2. J. Munroe, **Z. Perzan** and W. Amidon (2016). Cave sediments constrain Pleistocene advance of the Laurentide ice sheet in the Champlain Valley, *Journal of Quaternary Science*. [\[link\]](#)
1. A. Schroth, C. Giles, P. Isles, Y. Xu, **Z. Perzan** and G. Druschel (2015). Dynamic coupling of iron, manganese and phosphorus behavior in water and sediment of shallow ice-covered eutrophic lakes, *Environmental Science & Technology*. [\[link\]](#)

INVITED TALKS

Basin Characterization Program, <i>California Dept. of Water Resources</i>	2026
Association of Environmental and Engineering Geologists, <i>Southern Nevada AEG</i>	2025
Managed Aquifer Recharge Symposium, <i>CANVAS</i>	2025
Hydrogeophysics Session, <i>AGU Fall Meeting</i>	2024
Pardee Keynote Symposium, <i>GSA Annual Meeting</i>	2024
Groundwater Recharge Session, <i>GSA Annual Meeting</i>	2024

Environmental Geophysics Seminar, <i>Stanford University</i>	2024
Earth Science Seminar, <i>Dartmouth College</i>	2024

MAJOR RESEARCH FUNDING

<i>EPSCoR Research Fellows: Capturing Transient Metal Cycling in Soils and Groundwater through Real-Time Electrochemical Measurements</i> National Science Foundation (\$242,791) Z. Perzan (PI)	2027 – 2028
<i>Assessing Phosphorus Storage and Release from the Las Vegas Wash to Lake Mead</i> U.S. Bureau of Reclamation (\$302,014) Z. Perzan (PI) and T. Ashley	2026 – 2029
<i>Uranium cycling during episodic oxygenation events in floodplain soils</i> University of Nevada Las Vegas Z. Perzan (PI), R. Murali, and H. Xu	2026 – 2029
<i>Influence of Aridification on Future Phosphorus Storage and Mobility in the Las Vegas Wash</i> U.S. Geological Survey (\$48,000) Z. Perzan (PI) and A. Sloat	2026
<i>Understanding the impacts of managed aquifer recharge on the water balance in intensively managed basins</i> National Science Foundation (\$290,925) Z. Perzan (PI)	2025 – 2028
<i>Graduate experiences for future leaders: Building workforce capacity in resource management and mining</i> National Science Foundation (\$2,385,000) C. Tyler (PI), Z. Perzan , A. Martin, H. Bauser, and Y. Hu	2025 – 2028
<i>Monitoring aquifer recharge dynamics in the deep vadose zone using borehole nuclear magnetic resonance logging</i> U.S. Department of Agriculture (\$200,000) Z. Perzan (PI)	2024 – 2026
<i>Identifying controls on focused groundwater recharge across scales</i> National Science Foundation (\$180,000) Z. Perzan (PI; awarded but declined)	2023

TEACHING

Instructor of Record

Contaminant Hydrogeology (GEOL 475/675) University of Nevada, Las Vegas	Spring 2025, Spring 2026
Uncertainty Quantification in Hydrology (GEOL 767) University of Nevada, Las Vegas	Fall 2025, Fall 2026

Seminar in Geoscience (GEOL 491) University of Nevada, Las Vegas	Spring 2025
Introduction to Geochemistry (GEOL 330) University of Nevada, Las Vegas	Fall 2024
Indigenous Environmental Justice (ESS 226) Stanford University	Spring 2022
<i>Teaching Assistant</i>	
Intro. to Environmental Science (ENVS 110) University of San Francisco	Spring 2022
Contaminant Hydrogeology (ESS 221) Stanford University	Winters 2019 – 2023
Elements of Oceanography (GEOL 161) Middlebury College	Fall 2013 – 2014
<i>Pedagogical Training</i>	
Culturally Relevant & Responsive Teaching Fellow, <i>UNLV</i>	2025
Preparing Future Professors Program, <i>University of San Francisco</i>	2022
Designing a Learning-centered Syllabus Workshop, <i>Stanford University</i>	2022
Universal Design for Learning Workshop, <i>Stanford University</i>	2022
National Environmental Justice Education and Teaching Workshop	2021
Equitable Assessments Workshop, <i>Stanford University</i>	2021

MENTORSHIP

PhD students:

Đường (Kenny) Truong , Geoscience, University of Nevada, Las Vegas	2026 –
Babak Ghazi , Geoscience, University of Nevada, Las Vegas	2025 –

MS students:

Piper Irvin , Geoscience, University of Nevada, Las Vegas	2025 –
Brittany Meyers , Geoscience, University of Nevada, Las Vegas	2025 –
Sam Poppenhouse , Geoscience, University of Nevada, Las Vegas	2024 –
Timothy Dai , Computer Science, Stanford University	2022 – 2024
Ziyan Wu , Environmental Engineering, Stanford University	2019 – 2021

Undergraduate students:

Gisell Ponce , Geoscience, University of Nevada, Las Vegas	2026 –
Chanel Koh , Computer Science, University of Nevada, Reno	2024 –
Marc Berghouse , Earth System Science, Stanford University	2018 – 2021
Bailey Lewis , Environmental Science, Central Wyoming College	2019 – 2020
Cassie Weed , Environmental Science, Central Wyoming College	2019 – 2020
Diana Velazquez , Earth System Science, Stanford University	2019
George Sims , Environmental Science, Central Wyoming College	2018 – 2019
Dustin Proctor , Environmental Science, Central Wyoming College	2018 – 2019

SERVICE AND OUTREACH

Secretary, USDA Soil Physics Multistate Research Group USDA Multistate Research Project W5188	2026 –
Graduate Coordinator, Online Water Resources MS Program University of Nevada, Las Vegas	2024 –
Instructor, Reactive Transport Workshop NSF Reactive Transport Research Coordination Network	2024 – 2025
Panel Reviewer National Science Foundation	2025
Ad Hoc Proposal Reviewer Cascade Climate	2024
K-12 Outreach Events, Northern Arapaho Tribe Wind River Indian Reservation, Wyoming	2019 – 2023
Small Business Research Advisor Quantitative BioSciences, Inc.	2019 – 2023
Session Chair, Groundwater-Surface Water Interactions American Geophysical Union Fall Meeting	2022
Ad Hoc Proposal Reviewer DOE Biological and Environmental Research Program	2022
Reviewer for: Water Resources Research, Hydrology and Earth System Sciences, Environmental Science & Technology, Environmental Modelling and Software, Hydrological Processes, Journal of Hydrology, Science of the Total Environment, Agricultural Water Management	